

D S.

Arrays & Strings

LL

Stacks & Queue

B.Trees, BST

Graph

HashMap

heap

Exclusive to CPP & C

Pointers

sar dard

mazedaar

in C++

VVV Imp

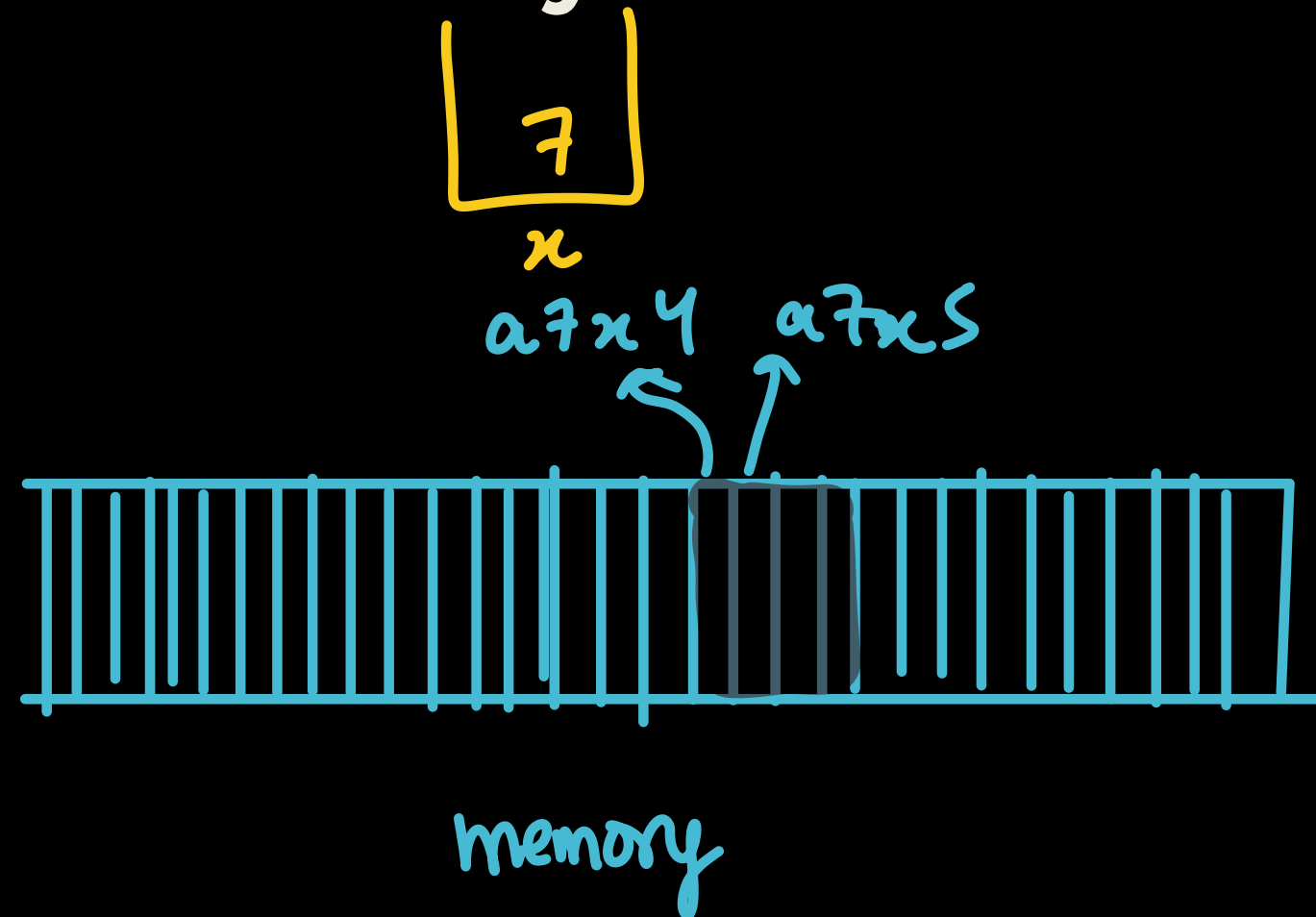
Why?

(Lecture 15)

Revisiting Variables

What a variable actually is ?

```
int x = 7 ;
```



Address Operator

Understanding address-of operator

A way to store addresses

Introduction to Pointers

```
int main(){  
    ✓ int a = 5;  
    ✓ int* ptr = &a;  
    ✓ cout<<&a<<endl;  
    ✓ cout<<ptr<<endl;  
    ✓ cout<<&ptr<<endl;  
}
```

$\boxed{5}$
a
x25

$\boxed{x25}$
ptr
y15

Output

. x25
x25
y15

Pointers

```
data_type * pointer_name;
```

```
int* x ;
```

```
float* y ,
```

```
char* ptr;
```

Why such **Syntax** ?

Why not a simple syntax such as:

```
pointer pointer_name;
```

Dereference Operator

```
✓ int x = 139;  
✓ int* p = &x;  
✓ cout << *p ;
```

dereference
operator

139
x
a240

a240
p
a900

Dereference Operator (*)

```
✓ int x = 139;  
✓ int* p = &x;  
✓ cout << x << endl;  
✓ *p = 10; // x=10  
✓ cout << x ;
```

$\begin{bmatrix} 10 \\ 139 \end{bmatrix}$
x
a510

$\begin{bmatrix} a510 \end{bmatrix}$
p
a290

Output
· 139
10

Dereference Operator

```
void change(int* ptr){  
    ✓ *ptr = 20;  
}  
  
int main(){  
    ✓ int x = 10;  
    ✓ change(&x);  
    ✓ cout<<x<<endl;  
}
```

a200
ptr
b100

20
~~10~~
x
a200

Output

20

Ques: Write a program to calculate sum of two numbers using pointers.

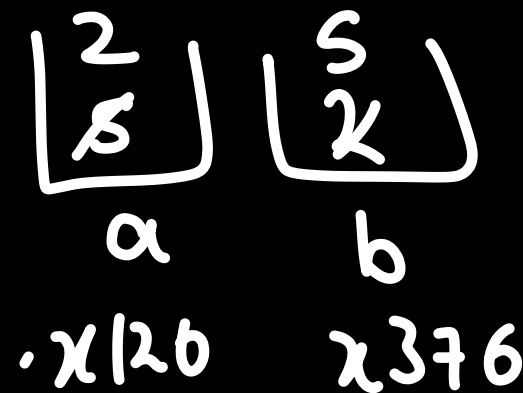
Pointers Syntax Problem

```
int x = 50, y = 91;  
int* p1 = &x, p2 = &y; → error
```

Pass by Value and Pass by Reference

Swap Function

```
✓ void swap(int* a, int* b){  
    ✓ int temp = *a;  
    ✓ *a = *b;  
    ✓ *b = temp;  
}  
  
int main(){  
    ✓ int a = 5, b = 2;  
    ✓ swap(&a, &b);  
    ✓ cout << a << " " << b << endl;  
}
```



Output
2 5

Pass by **Reference** (using alias)

Alias names using & operator

Already done

Pointer Arithmetic

Increment & Decrement

Will be done in

Arrays

Pointer Arithmetic

The dependence of addition and subtraction to pointers on the data type

Ques: Predict the output

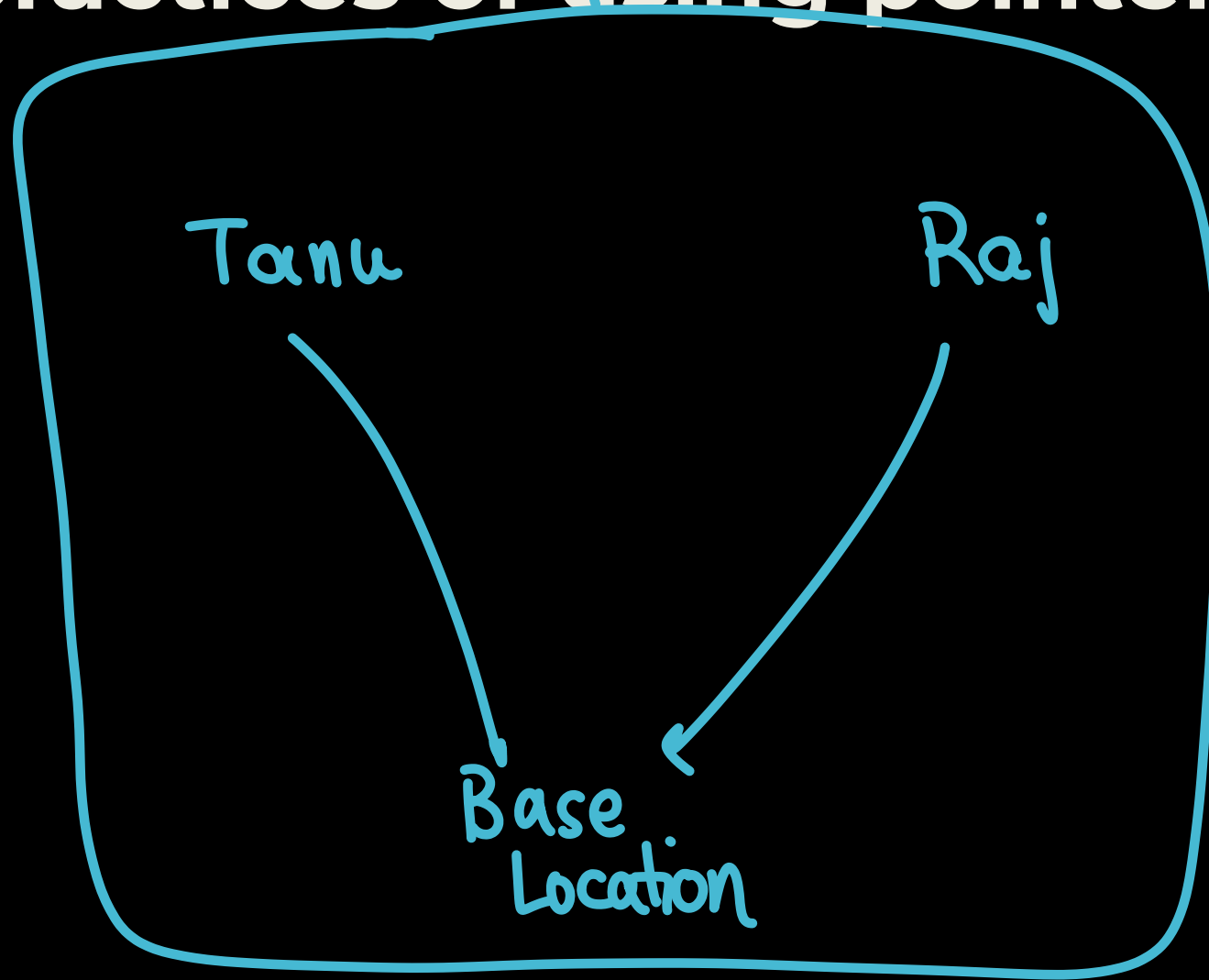
```
int a = 15;  
int *ptr = &a;  
int b = ++*ptr;  
  
cout << a << " " << b;  
  
// Assume the address of a is 1000.
```


Ques: Write a function to find out the first and last digit of a number without returning anything.

Null Pointer

शून्य

Good practices of using pointers



Double Pointers

used to store address of single pointer

Double Pointers

```
int x = 21;  
int* ptr = &x;  
int** p = &ptr;  
  
cout << x << endl;  
cout << *ptr << endl;  
cout << **p << endl;
```

This was the **LAST** class of C++

Accha College → Jee (PCM)

Acchi Placement → DSA + DEV
or

DS

or

AI-ML

just 12th pass → Noida
Meetup